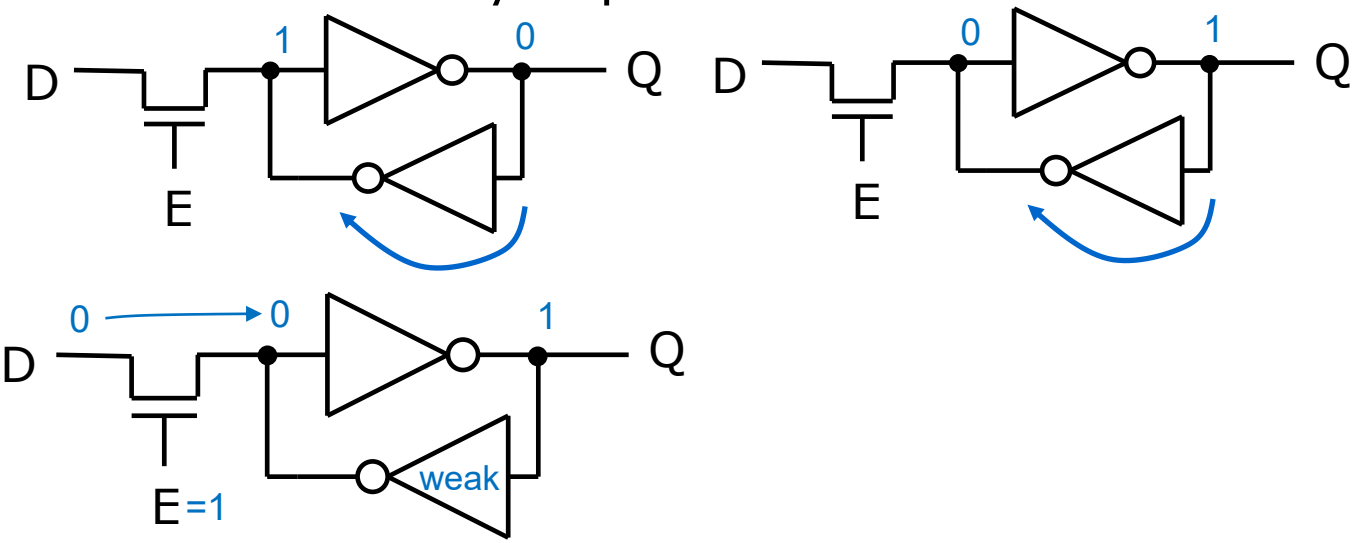
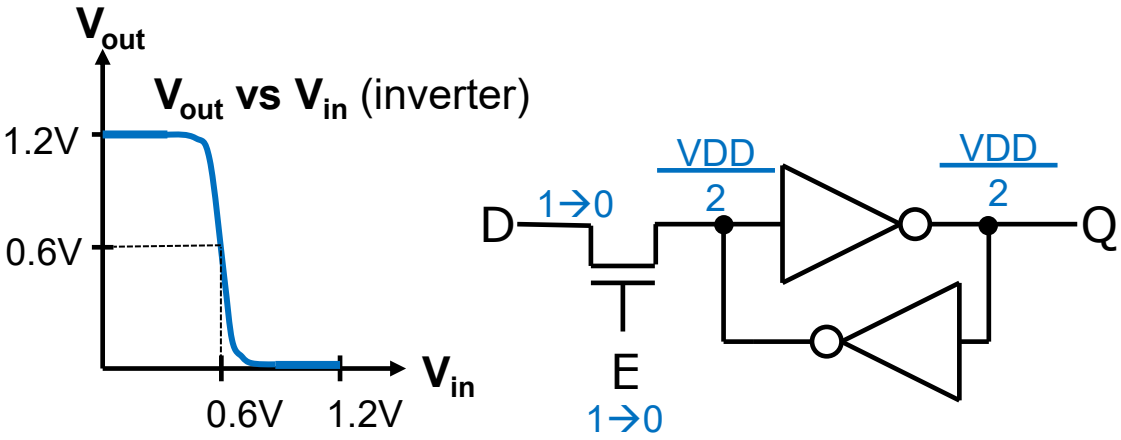
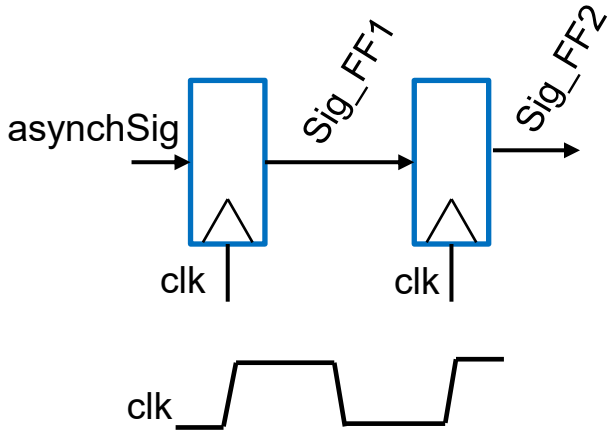
Metastability Recap

- Consider the memory loop in a latch



- May become metastable if signal is captured during transition





Exercise 03 Synchronizing Edge Detector

Submit a **structural** Verilog file that models a module that can synchronize a signal to our system clock (**clk**) and output a signal that is high for 1 clock cycle when that signal has a rising edge. Assume a D-FF is already defined that has an interface of:(D, clk,Q,PRN) and a module name of **dff**.

- What do you need to solve metastability?
- What do you need to detect a rising edge?

- Name your file **synch_detect.sv** and ensure the module name is also **synch_detect**

- Your interface should have:

  asynch_sig_in → input to synchronize and detect rising edge on
  clk → clock signal
  rst_n → global asynch active low reset (initialize the circuit) (**pre**set flops in this case)
  rise_edge → output of your block that is high for 1 clock cycle after rising edge occurs on

Exercise 03 Synchronizing Edge Detector

Testbench

- Download the provided **dff.sv** and **synch_detect_tb.sv**
- Create a Modelsim project and test your design. The test bench is self-checking and should give a happy message in the console window if your DUT is good.

→ **Submit synch_detect.sv**

- Once you are done, there are a few extra questions on the submissions page for you to answer.